

States of matter and changes of state

Learning outcomes

By the end of this section you should be able to:

- Distinguish the different states of matter.
- Use state symbols in chemical reactions (s, l, g, aq).

All matter that exists today was created in the Big Bang that occurred approximately 13.8 billion years ago. The Big Bang theory states that the universe started from a small dot about the size of a period on this page that underwent a sudden expansion. About 380 000 years after this expansion, the first element (hydrogen) began to form. Over the next several billion years stars and galaxies were formed, giving rise to the universe that we know today. Isn't it amazing to think that the matter that makes up everything in our universe now is the same matter produced in the Big Bang nearly 14 billion years ago?

Nature of Science

Aspect: Experiments

Experiments can often lead to new discoveries. Scientists have recently discovered a new state of matter, known as liquid glass  (<https://www.techexplorist.com/new-state-matter-liquid-glass/37171/>). This new state of matter has properties between that of a solid and a colloid (a type of gel) and could have important implications for the future development of liquid crystal displays (LCDs).

States of matter

The three states of matter that we experience in our daily lives are solid, liquid and gas. A fourth state of matter – plasma – which is an ionised gas, exists mainly in space. In this section we will focus on the three common states of matter which

are solid, liquid and gas. The three states differ in characteristics such as their shape or the ability to flow and in properties such as density and compressibility. To understand the properties of the states of matter we apply the kinetic molecular theory (KMT).

According to this theory:

- All matter is made up of small particles.
- These particles all have kinetic energy (the energy of motion) which causes the particles to constantly move.
- The amount of kinetic energy is proportional to the temperature of the substance; therefore, the particles have greater motion at higher temperatures (straight line motion) and lesser motion at lower temperatures (vibrational motion).
- Collisions between particles are elastic, which means no loss in kinetic energy.

Using these assumptions, we can explain the properties of the different states of matter.

Making connections

Why are some substances solid while others are fluid (liquid) under standard conditions? This will be explored in more detail in subtopic S2.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43125/>).

The particle models of solids, liquids and gases are shown in **Figure 1**. In a solid, the particles are held in fixed positions but can vibrate around a fixed point. In a liquid, the particles are more spread out but are still attracted to each other. And in a gas, the particles are very spread out and move randomly in straight lines.

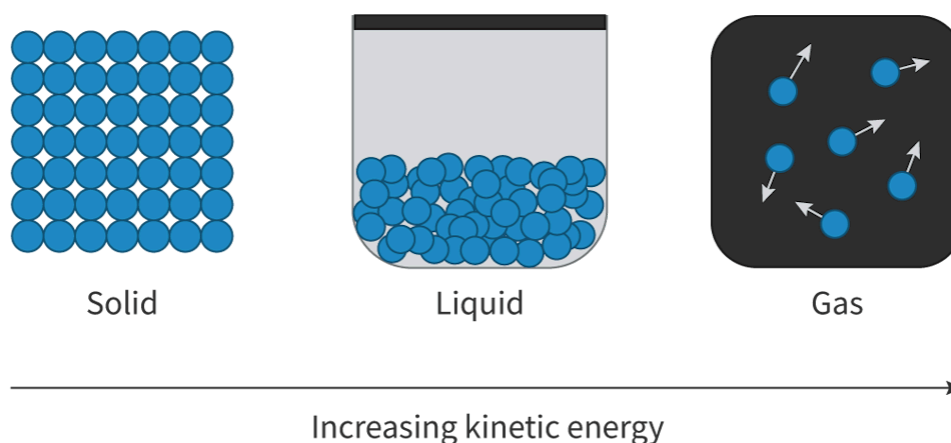


Figure 1. Particle models of solid, liquids and gases.

 More information for figure 1

This diagram illustrates particle models for solids, liquids, and gases. On the left, the solid model consists of small circles organized in a tightly packed square grid, representing particles in fixed positions. In the center, the liquid model is depicted within a beaker, where the particles are loosely arranged and overlapping at the bottom, allowing for movement while staying close together. On the right, the gas model shows particles spread out and distant from each other, with arrows marking random motion, indicating that they are free and rapidly moving. The diagram also implies increasing kinetic energy from solid to gas.

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The properties of solids, liquids and gases are closely related to the arrangement of the particles in the substance.

- Solids, for example, cannot be compressed because the particles are already very close together with little space between them. There are strong forces of attraction that hold the particles together, so they have a fixed shape and a fixed volume. For the same reason, solids cannot flow.
- Liquids can flow because the particles are able to move more freely than in a solid. There are still forces of attraction between

the particles, but these are weaker than in a solid. Liquids do not have a fixed shape however and tend to take the same shape as the bottom of the container they are in. The particles in a liquid are still quite close together, therefore liquids cannot be compressed.

- Gases do not have a fixed shape nor a fixed volume; they take the same shape as the container and their volume depends on the temperature and the pressure of the gas itself. The forces of attraction between the particles are the weakest among the three states. Gases are the only state of matter that can be compressed because the particles are far apart with lots of space between them.

In terms of kinetic energy, solids have the least amount of kinetic energy, and gases the most. This concept will be discussed in more detail in the next section.

Density

The density of a substance is its mass per unit volume. Substances with higher densities will feel 'heavier' compared to substances with lower densities (of the same volume). To calculate the density, d , of a substance, its mass, m , is divided by its volume, V .

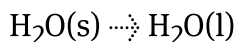
$$d = \frac{m}{v}$$

Solids tend to have higher densities than liquids which tend to have higher densities than gases. For example, air, which is a mixture of gases, has a density of 0.0018 g cm^{-3} . The density of water, for comparison, is 1.00 g cm^{-3} and the density of solid iron is 7.87 g cm^{-3} .

Changes of state

Changes of state (also called phase changes) occur when a substance changes from one physical state to another. Take for example, the melting of ice when left on a kitchen counter. The temperature of the surroundings is higher than the temperature of the ice which causes a movement of heat energy from the surroundings to the ice. After a period of time the ice has melted and become

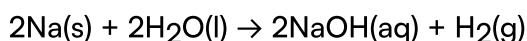
liquid water. This is a physical change because the water itself is chemically unchanged so it has only changed from solid water to liquid water. We can represent the melting of ice as follows:



Note the use of state symbols that show the physical state of the substances. The (s) means the substance is in the solid state and the (l) means the substance is in the liquid state. The other two state symbols commonly used are (g) for the gaseous state and (aq) for substances dissolved in aqueous solutions.

Study skills

The state symbols (s), (l), (g) and (aq) are also used when writing equations that represent chemical reactions. One such example that will be studied in section S3.1.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/group-1-and-17-elements-id-44694/>) is the reaction between sodium metal and water to produce sodium hydroxide and hydrogen gas:



In this chemical equation, we have all four state symbols used. Sodium is a solid (s), water is a liquid (l), sodium hydroxide forms an aqueous solution (aq) and hydrogen is a gas (g). It is worth noting that 'aqueous' means a homogeneous mixture made by dissolving the substance in water, also referred to as a solution. It is good practice to include state symbols when writing chemical equations, so make sure that you look out for them in future sections.

Physical changes are reversible processes meaning that they can go in either direction. For example, ice can melt to form water and water can freeze again to become ice. Other changes of state are less common in our everyday lives such as sublimation and deposition.

- Sublimation is the change of state directly from a solid to a gas with no liquid phase. Sublimation can be observed when dry ice (solid carbon dioxide) sublimates from a solid directly to a gas.
- Deposition is the reverse process, the direct change from a gas to solid with no liquid phase. An example is the formation of frost in a freezer as the moisture in the air forms solid ice.

It is also worth noting the difference between evaporation and boiling which are collectively known as vaporisation.

- Evaporation is a change of state from liquid to gas and takes place only at the surface of the liquid. It can occur at temperatures below the boiling point of the liquid.
- Boiling occurs at a specific temperature and is a change of state from liquid to gas throughout the liquid. Bubbles of gas are formed within the liquid, not only at the surface.

The changes of state are summarised in **Figure 2**.

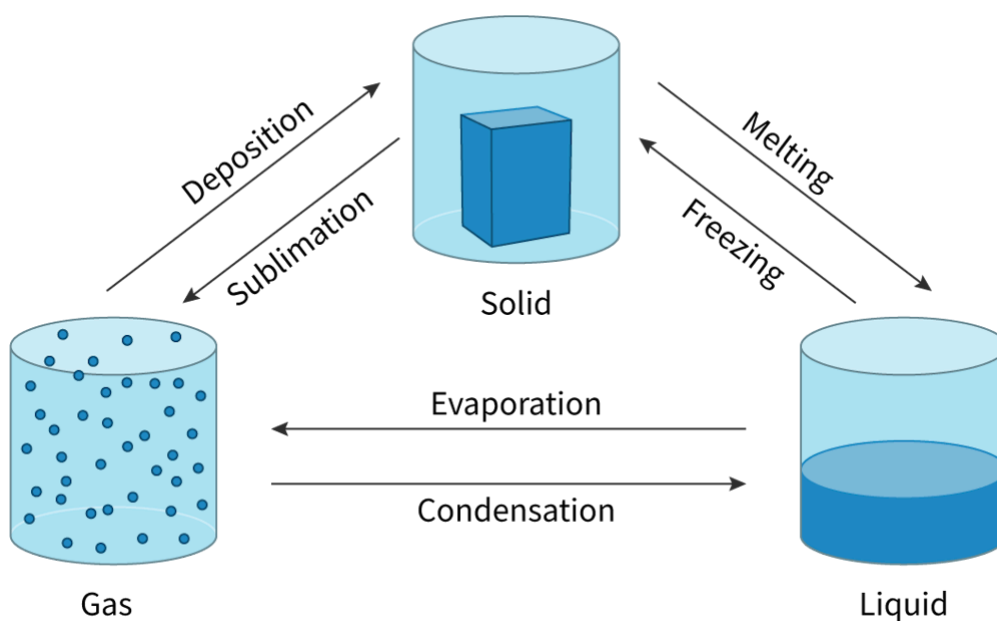


Figure 2. The states of matter of water together with the changes of state.

 More information for figure 2

A diagram illustrating the particle models for solids, liquids, and gases within beakers, accompanied by arrows indicating the changes of state. The solid is depicted by a cuboid shape inside a beaker, the liquid by a flat horizontal line in a beaker, and the gas by floating circles.

Arrows between these models show six state changes: - Solid to Liquid: Melting. - Liquid to Solid: Freezing. - Liquid to Gas: Evaporation. - Gas to Liquid: Condensation. - Gas to Solid: Deposition. - Solid to Gas: Sublimation.

Solid is in the top center, gas on the left, and liquid on the right, forming a triangular flow of transitions among them.

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Theory of Knowledge

Water is one of the few substances that we can observe all changes in state in our everyday life. To what extent do you rely on your experience of macroscopic changes to understand what is happening on a microscopic level?

Use the activity in the next section (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/activity-states-of-matter-id-43138/>) to investigate the states of matter.